

broadcast. When data transmission is directly from sensor node to sink node it is called unicast.

Protocol operation based. These can be route-discovery based, data-delivery mode based, node-location based and QoS based. Route-discovery based protocols are classified as proactive, reactive and hybrid. Data-delivery based routing protocols are classified as continuous, event based and query based. Node-location based protocols are classified as flooding and greeding. The QoS based classification depends on network conditions, traffic classes and real-time streaming.

Application based. A routing protocol selected for one kind of application will not be suitable for other types.

Coverage based. Coverage can be considered a quality of service (QoS) which guarantees that all points in the region of interest (RoI) are covered at least by one sensor. The coverage can be further classified as: (a) area coverage, (b) barrier coverage, whose main objective is to detect the intrusion of any object in the RoI and to minimise the probability of undetected intrusion of an object, (c) point coverage, when the goal is to cover set of points in the RoI, and (d) path coverage, to cover a particular path that an object follows to reach from one point to another.

Connectivity and coverage problems are caused by the limited communication and sensing range. There are basically three main reasons that cause coverage problems: random deployment, limited sensing range and not enough sensors to cover the whole RoI. Strategies used in solving the coverage problem can be divided into three categories: force based, grid based and computational-geometry based.

Ratio of area covered to the area of RoI is called coverage percentage. Coverage is estimated as ratio of grid points covered to total number of grid points in RoI. Computational geometry is based on coverage optimisation.

Hardware presentation: ZigBee, XBee, SquidBee

ZigBee is a standard wireless network

protocol designed for low-rate control networks. Its applications include automation networks, security systems, industrial control networks, remote meter reading and PC peripherals connectivity. ZigBee is the name of a specification for a suite of high-level communication protocols using small, low-power digital radios based on IEEE 802.15.4 standard for wireless personal area networks (WPANs), such as wireless headphones connected to cell phones via short-range radio.

ZigBee has a much wider range as compared to Bluetooth. It covers the network layer, application framework and application profiles, which guarantees interoperability between equipment from different vendors. In addition, the ZigBee network layer specification defines networking topologies like peer-to-peer, cluster or mesh network configurations. The cluster and mesh networks are also known as multi-hop networks, whereas star network is called single-hop network. ZigBee network can activate (go from sleep to active mode) in 30 milliseconds or less, the latency can be very low and devices can be very responsive. Because ZigBee can 'sleep' most of the time, average power consumption can be very low.

ZigBee's use of IEEE 802.15.4 PHY & MAC standards allows networks to handle any number of devices. ZigBee's protocol code stack is estimated to be about one-fourth of Bluetooth's or 802.11's. ZigBee may secure messages transmitted over a single hop using secured MAC data frames, but for multi-hop messaging it relies upon upper layers (such as NWK layer) for security. The ZigBee alliance is not pushing a technology, rather it is providing a standardised base set of solutions for sensors and control systems.

There are several WSN hardware platforms available. One of them could be to base an Arduino board and XBee module. Arduino boards are available with ATmega 8, 16, ..., 328, and are equipped with a library for interfacing with ZigBee/XBee modules for dealing with analogue or digital

inputs and outputs. These are modules for easy programming (in C) of the microcontroller. This module can be interfaced with a PC using Visual Basic so that the parameters can be monitored.

Another WSN platform developed is SquidBee, which is an open source device in which each SquidBee Module takes environmental parameters like temperature and humidity with its sensors and sends them wirelessly through the SquidBee network using ZigBee protocol.

ZigBee arrived in 2.4GHz band 50 years ago, joining the well-established Bluetooth and Wi-Fi technologies. Inseton combines wired power line and wireless communication to form mesh networks for the devices to communicate when in proximity. If one device fails the others can take over.

XBee and XBee PRO ZB are the most advanced ZigBee modules available, and are ideal for deployment in ZigBee networks. XBee-embedded RF modules share a common hardware footprint and are modelled after a common software application programming interface (API). Once deployed in an application, an original equipment manufacturer (OEM) can rapidly change from one protocol to another with minimal time and development risk.

With advanced mesh networking functionality, XBee PRO ZB modules improve data traffic management, allow for greater node density and provide OEMs with the ability to change firmware remotely with over-the-air updates. XBee-embedded modules are available with different protocols to suit a variety of applications and network topologies. Supported protocols include IEEE 802.15.4, the Zigbee PRO feature set, proprietary long range and Digimesh.

There are many simulator softwares for WSN including NS-2, SENSE, NETSIM, OPNET and OMNeT+ which can be used to analyse data packet delivery, probability of packet being discarded and other parameters.

To be continued next month

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